

$$p = \begin{cases} \frac{e^{-x/\theta}}{\theta}, & x \geq 0 \\ 0, & x < 0 \end{cases} \quad \theta > 0$$

T3
 $\vec{X}_n$ - выборка, $n=3$

$$\tilde{\theta}_1 = \bar{x}$$

$$\tilde{\theta}_2 = X_{(2)}$$

a) проверить на непереносимость

$$M\tilde{\theta} = \theta$$

1) $M\tilde{\theta}_1 = M\left(\frac{1}{n} \sum_{i=1}^n x_i\right) = M_{\bar{x}} = \theta \Rightarrow \tilde{\theta}_1$ - несмещенная оценка

найдём $M_{\bar{x}} = \int_{-\infty}^{+\infty} x p(x) dx = \frac{1}{\theta} \int_0^{+\infty} x e^{-x/\theta} dx = - \int_0^{+\infty} x d e^{-x/\theta} = - \frac{x e^{-x/\theta}}{\theta} \Big|_0^{+\infty} + \int_0^{+\infty} e^{-x/\theta} dx = - \theta e^{-x/\theta} \Big|_0^{+\infty} = \theta$

2) $X_{(2)} \sim \sum_{i=2}^3 C_3^i (1 - e^{-x/\theta})^i e^{-x/\theta \cdot (3-i)} = 3(1 - e^{-x/\theta})^2 e^{-x/\theta} + (1 - e^{-x/\theta})^3 = (1 - e^{-x/\theta})^2 (2e^{-x/\theta} + 1) = F(x)$

$$p(x) = F'(x) = 2(1 - e^{-x/\theta}) \cdot e^{-x/\theta} \left(\frac{1}{\theta}\right) (2e^{-x/\theta} + 1) + (1 - e^{-x/\theta})^2 \cdot 2 \cdot e^{-x/\theta} \cdot \left(-\frac{1}{\theta}\right) =$$

$$= \frac{2}{\theta} (1 - e^{-x/\theta}) e^{-x/\theta} [2e^{-x/\theta} + 1 - 1 + e^{-x/\theta}] = \frac{6}{\theta} (1 - e^{-x/\theta}) e^{-2x/\theta} = \frac{6}{\theta} \left[e^{-2x/\theta} - e^{-3x/\theta} \right]$$

$$M\tilde{\theta}_2 = \frac{6}{\theta} \left[\int_0^{+\infty} x e^{-2x/\theta} dx - \int_0^{+\infty} x e^{-3x/\theta} dx \right] = \frac{6}{\theta} \left[-\frac{\theta}{2} \int_0^{+\infty} x d e^{-2x/\theta} + \frac{\theta}{3} \int_0^{+\infty} x d e^{-3x/\theta} \right] = 6 \left[-\frac{1}{2} x e^{-2x/\theta} \Big|_0^{+\infty} + \frac{1}{2} \int_0^{+\infty} e^{-2x/\theta} dx + \frac{1}{3} x e^{-3x/\theta} \Big|_0^{+\infty} - \frac{1}{3} \int_0^{+\infty} e^{-3x/\theta} dx \right] =$$

$$= 6 \left[\frac{\theta}{4} - \frac{\theta}{9} \right] = \frac{3}{2}\theta - \frac{2}{3}\theta = \frac{5}{6}\theta \Rightarrow \tilde{\theta}_2$$
 - смещенная

$$\tilde{\tilde{\theta}}_2 = \frac{6}{5}\tilde{\theta}_2$$
 - исправленная несмещенная оценка

b) оценить оптимальность:

1) $\varphi\tilde{\theta}_1 = \frac{\varphi_1}{n} = \frac{\theta^2}{n} \Big|_{n=3} = \frac{\theta^2}{3}$

найдём $\varphi_{\tilde{\theta}_2}$: $M_{\tilde{\theta}_2}^2 = \frac{1}{\theta} \int_0^{+\infty} x^2 e^{-x/\theta} dx = -x \frac{e^{-x/\theta}}{\theta} \Big|_0^{+\infty} + 2 \int_0^{+\infty} x e^{-x/\theta} dx = 2\theta^2$

$\varphi_{\tilde{\theta}_2} = M_{\tilde{\theta}_2}^2 - (M_{\tilde{\theta}_2})^2 = 2\theta^2 - \theta^2 = \theta^2$

2) найдём $M\tilde{\tilde{\theta}}_2$

$$M\tilde{\tilde{\theta}}_2 = \frac{36 \cdot 6}{25\theta} \int_0^{+\infty} x^2 \cdot \frac{6}{\theta} [e^{-2x/\theta} - e^{-3x/\theta}] dx = \frac{36 \cdot 6}{25\theta} \left\{ \int_0^{+\infty} x^2 e^{-2x/\theta} dx - \int_0^{+\infty} x^2 e^{-3x/\theta} dx \right\} = \frac{36 \cdot 6}{25\theta} \left\{ \int_0^{+\infty} x^2 e^{-ax} dx = \frac{2}{a^3} \int_0^{+\infty} x e^{-ax} dx = \frac{2}{a^3} \right\} \ominus$$

$$= \frac{36 \cdot 6}{25\theta} \left\{ \frac{1}{4}\theta^3 - \frac{2}{27}\theta^3 \right\} = \theta^2 \frac{38}{25}$$

$$\varphi\tilde{\tilde{\theta}}_2 = \theta^2 \left(\frac{38}{25} - 1 \right) = \frac{13}{25}\theta^2$$

$$\frac{13}{25}\theta^2 > \frac{1}{3}\theta^2 \Rightarrow \tilde{\theta}_1$$
 - эффективнее, чем $\tilde{\tilde{\theta}}_2$

с) 1) проверим регулярность модели

1) $\frac{1}{\theta} e^{-x/\theta} \in C'(0, +\infty)$ ✓

2) $\int_0^{+\infty} \frac{\partial p}{\partial \theta} dx = \int_0^{+\infty} \left(-\frac{1}{\theta^2} e^{-x/\theta} + \frac{1}{\theta} e^{-x/\theta} \cdot \frac{x}{\theta^2} \right) dx = -\frac{1}{\theta^2} + \frac{1}{\theta^3} \int_0^{+\infty} x e^{-x/\theta} dx = -\frac{1}{\theta^2} + \frac{1}{\theta^2} = 0$ ✓

3) $I(\theta) = M \left[\frac{\partial \ln p(x, \theta)}{\partial \theta} \right]^2 = \int_0^{+\infty} \left(\frac{-\ln \theta - x/\theta}{\theta} \right)^2 \frac{1}{\theta} \exp(-\frac{x}{\theta}) dx = \int_0^{+\infty} \left(-\frac{1}{\theta} + \frac{x}{\theta^2} \right) \frac{1}{\theta} e^{-x/\theta} dx = \int_0^{+\infty} \left(\frac{1}{\theta^3} e^{-x/\theta} + \frac{x^2 e^{-x/\theta}}{\theta^5} - \frac{2x e^{-x/\theta}}{\theta^3} \right) dx$

$= \frac{\theta}{\theta^3} + \frac{2}{\theta^5} \theta^3 - \frac{2}{\theta^3} \theta = \frac{1}{\theta^2} \in C_\theta(0, +\infty)$ и > 0 на $(0, +\infty)$ ✓

$\Rightarrow$ модель регулярна

2) проверим на регулярность оценок.

• $\vartheta \tilde{\theta}_1 = \frac{\theta^2}{3}$ - оцр на $\forall$ компакте $\vartheta > 0$ и нелинейная $\Rightarrow \tilde{\theta}_1$ - регулярная оцска

• $\vartheta \tilde{\theta}_2 = \frac{13}{25} \theta^2$ - оцр на $\forall$ компакте $\vartheta > 0$ и нелинейная $\Rightarrow \tilde{\theta}_2$ - регулярная оцска

$\Rightarrow$ оцска регулярна

3) Вспомогательные неравенства Крамера-Рассо.

1) $\vartheta \tilde{\theta}_1 = \frac{\theta^2}{3} = \frac{\theta^2}{3} \rightarrow$ эффективная оцска

2) $\vartheta \tilde{\theta}_2 = \frac{13}{25} \theta^2 \geq \frac{\theta^2}{3} \rightarrow$ ничем связать не можем

✓

$g \sim p(x) = \alpha \{(-1, 1) \setminus \{0\}\} + \beta \{0\} + \gamma \{2\}$

$\int_{-\infty}^{+\infty} p(x) dx = \int_{-1}^1 \alpha dx + 2\beta = 2\alpha + 2\beta = 1 \Rightarrow \alpha + \beta = \frac{1}{2}$
 $\int x p(x) dx = 2\gamma = 1 \Rightarrow \gamma = \frac{1}{2} - \theta$

$\theta \in (0, \frac{1}{2})$

$\Rightarrow g \sim p(x) = \theta \{(-1, 1) \setminus \{0\}\} + (\frac{1}{2} - \theta) \{0\} + (\frac{1}{2} - \theta) \{2\}$ $\theta \in (0, \frac{1}{2})$

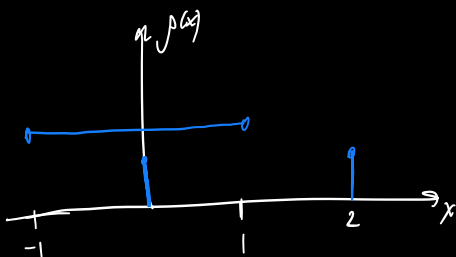
а) 1) $I \tilde{x}_n$ - выборка

$\lambda_1 = M \tilde{g} = \int_{-1}^1 x \theta dx + 0 \cdot (\frac{1}{2} - \theta) + 2(\frac{1}{2} - \theta) = 1 - 2\theta = \lambda_1(\theta)$

$\lambda_1(\theta) = \tilde{\lambda}_1 = \bar{x}$

$1 - 2\theta = \bar{x} \Rightarrow \tilde{\theta}_1 = \frac{1 - \bar{x}}{2}$ ОММ

2)



I_m - каи-бо $\{0\}$ и $\{2\}$ в $\tilde{x}_n$

$L(\theta) = \prod_{i=1}^n p(x_i, \theta) = \theta^{n-m} \cdot (\frac{1}{2} - \theta)^m$

$\ln L(\theta) = (n-m) \ln \theta + m \ln (\frac{1}{2} - \theta)$

$(\ln L)' = \frac{n-m}{\theta} - \frac{m}{1/2 - \theta} = 0$

$(n-m)(1/2 - \theta) - m\theta = \frac{n-m}{2} - \theta(n-m) + m\theta - m\theta \Rightarrow \tilde{\theta} = \frac{n-m}{2n} = \frac{1}{2} - \frac{1}{2} = \tilde{\theta}_1$ ОМП

$(\ln L)'' = -\frac{n-m}{\theta^2} - \frac{m}{(1/2 - \theta)^2} < 0 \Rightarrow$ т. max

1) $\tilde{\theta}_1 = \frac{1-\bar{x}}{2}$
 $M\tilde{\theta}_1 = M\left[\frac{1}{2} - \frac{\bar{x}}{2}\right] = \frac{1}{2} - \frac{M\bar{x}}{2} = \frac{1}{2} - \frac{1-2\theta}{2} = \frac{1}{2} - \frac{1}{2} + \theta = \theta$ — несмещенная
 $D\tilde{\theta}_1 = \frac{1}{4} \frac{D\bar{x}}{n} \xrightarrow{n \rightarrow \infty} 0$ — состоятельная

2) $\tilde{\theta}_2 = \frac{1}{2}(1-\bar{y})$
 $M\tilde{\theta}_2 = \frac{1}{2} - \frac{1}{2}M\bar{y} = \frac{1}{2} - \left(\frac{1}{2} - \theta\right) = \theta$ — несмещенная
 $D\tilde{\theta}_2 = \frac{1}{4} \cdot D\bar{y} = \frac{\theta(1/2-\theta)}{n}$

3) 1) проверим регулярность модели

1) θ и $\frac{1-2\theta}{2} \in C_\theta(0, 1/2)$ (здесь нужно проверить по нстр, т.к. $p(x)$ — производная $F(x)$)

2) $\int d \frac{\partial}{\partial \theta} F = \int_{-1}^1 \frac{\partial \theta}{\partial \theta} dx + \frac{\partial(1/2-\theta)}{\partial \theta} \cdot 0 + \frac{\partial(1/2-\theta)}{\partial \theta} \cdot 2 = 2 - 2 = 0$

3) $I(\theta) = \int_{-1}^1 \left(\frac{\partial \ln \theta}{\partial \theta}\right)^2 \cdot \theta dx + 2 \left(\frac{\partial \ln(1/2-\theta)}{\partial \theta}\right)^2 (1/2-\theta) = \frac{2}{\theta} + \frac{2(1/2-\theta)}{(1/2-\theta)^2} = \frac{2}{\theta} + \frac{4}{1-2\theta} \in C_\theta(0, 2)$ и > 0

→ модель регулярная

2.1) регулярность $\tilde{\theta}_1$:

$M\tilde{\theta}_1^2 = \int_{-1}^1 x^2 \theta dx + 4 \left(\frac{1}{2} - \theta\right) = \frac{x^3 \theta}{3} \Big|_{-1}^1 + 2 - 4\theta = \frac{2\theta}{3} + 2 - 4\theta = -\frac{10}{3}\theta + 2$

$D\tilde{\theta}_1 = \frac{-\frac{10}{3}\theta + 2 - (-2\theta)^2}{4n} = \frac{-\frac{10}{3}\theta + 2 - 1 - 4\theta^2 + 4\theta}{4n} = \frac{\frac{2}{3}\theta + 1 - 4\theta^2}{4n}$

$D\tilde{\theta}_1 = \frac{1 + \frac{2}{3}\theta - 4\theta^2}{4n}$ — окр на $\forall$ кантате $y(0, 1/2) \Rightarrow \tilde{\theta}_1$ — регулярная по достаточному условию

2.2) регулярность $\tilde{\theta}_2$

$\frac{(1/2-\theta)\theta}{n}$ окр. на $\forall$ кантате $y(0, 1/2) \Rightarrow \tilde{\theta}_2$ — регулярная по достаточному условию

3) Неравенство Крамера-Рэо.

1) $D\tilde{\theta}_1 = \frac{1 + \frac{2}{3}\theta - 4\theta^2}{4n} \neq \frac{1}{n \left(\frac{2}{\theta} + \frac{4}{1-2\theta}\right)} = \frac{\theta(1-2\theta)}{2n}$

→ нечего сказать об эффективности

2) $D\tilde{\theta}_2 = \frac{(1-2\theta)\theta}{2n} = \frac{\theta(1-2\theta)}{2n} \rightarrow$ эффективная //